

Inversions at inverted repeats and the double-stranded fibre

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24 September 2026

Abstract

Two circular sequences with the same k -mer spectrum differ by transpositions of segments bounded by repeated words: this is the theorem of Ukkonen and Pevzner. Double-stranded data do not see the spectrum of one strand, only the sum of the spectra of both strands. The corresponding set of sequences is the double-stranded fibre $\text{DS}(S, k)$ counted in Bio 8, 12 and 13. Bio 8 identified the move that the second strand adds, the *inversion flip*. Write $T = PwZ'\text{rc}(w)R$ with w a k -mer. Replacing the segment $wZ'\text{rc}(w)$ by its reverse complement preserves the double-stranded spectrum. For ϕX174 at $k = 11, 12$ the flips alone connect the fibre. But flips alone cannot connect it when the genome has no inverted repeat. We add the lattice picture: a flip moves the point of Bio 8 by $-2(1 - \rho)z$, where z is a path from a vertex to its complement. Three results follow. (i) Transpositions and flips, together with global reverse complementation, generate the whole fibre in every case we can close. The cases are twenty-four random sequences, checked against exhaustive enumeration and against Bio 8's lattice sum, and ϕX174 at $k = 12$ and $k = 11$, where the closure of the genome is exactly $\#\text{DS} = 10$ and 864. Transpositions alone reach 2 and 33 sequences there, and adding reverse complementation reaches 4 and 66. Adding transpositions removes Bio 8's obstruction, and we state the general case as a conjecture. (ii) The *inversion lattice* is always all of Λ^- . The cycle part $(1 - \rho)Z_1$ has index

$$[\Lambda^- : (1 - \rho)Z_1] = \begin{cases} 2 & \text{if } D_k \text{ has no palindromic vertex and no self-complementary arc,} \\ 1 & \text{otherwise,} \end{cases}$$

by a Tate-cohomology computation, confirmed on 600 random covers. When the index is 2, the missing class is exactly an inversion vector. (iii) ϕX174 at $k = 10$ carries 31 inversions flanked by inverted repeats of length 10. Random walks of 20 000 moves stay inside the fibre of 31 925 753 246 212 reconstructions and visit about 5 000 of its lattice points. The inversion-group programme proposed for this paper (Artin groups, Garside normal forms) is set aside, and we say why.

Keywords. de Bruijn graph, reverse complement, inverted repeats, chromosomal inversions, Eulerian circuits, transpositions, Tate cohomology, genome assembly.

1 The question

Let S be a circular DNA sequence and k a word length. In the single-stranded world, the sequences with the same $(k + 1)$ -mer spectrum as S are the Eulerian circuits of one graph. Any

two of them are related by *transpositions*: exchanges of segments that start and end at repeated k -mers [5, 6]. A sequencer that cannot tell the strands apart sees only

$$\text{spec}(T) + \text{spec}(\text{rc } T) = \text{spec}(S) + \text{spec}(\text{rc } S),$$

and the set $\text{DS}(S, k)$ of circular T satisfying this is larger. Bio 8 wrote its size as a sum over the points of a lattice Λ^- in a box, Bio 12 counted the box, and Bio 13 computed $\#\text{DS}(\phi\text{X174}, 10) = 31\,925\,753\,246\,212$ [1, 2, 3]. A transposition never leaves the lattice point of T . So the question of this note is *which moves carry one lattice point to another*, and whether together with transpositions they connect the fibre.

The move itself is Bio 8's *inversion flip* [1, §7]: an inversion between two copies of an inverted repeat. It is an operation every geneticist knows. In the human genome it is the classical mechanism of recurrent inversions, via non-allelic homologous recombination between inverted low-copy repeats [9], and here it is the combinatorial content of the second strand. Bio 8 proved two things about it. The flips preserve the double-stranded spectrum, and a genome without inverted repeats admits none, so its flip graph is discrete. It also computed that for ϕX174 at $k = 12$ and $k = 11$ the flips alone connect all 10 and 864 reconstructions. This note adds three things. It combines flips with transpositions, which removes that obstruction and gives a general conjecture. It computes the lattice spanned by flip vectors (the Tate dichotomy). And it walks the fibre at $k = 10$.

2 Setting

We keep Bio 8's notation. D_k is the labelled reverse-complement double cover of S . Its vertices are the k -mers that occur at least twice in S and $\text{rc } S$ together. Its arcs, the *contents*, are the stretches between consecutive such k -mers. The involution ρ is reverse complementation, on vertices ($v \mapsto \bar{v}$) and on contents; a content $a: t \rightarrow h$ goes to $\rho a: \bar{h} \rightarrow \bar{t}$. A reconstruction $T \in \text{DS}(S, k)$ is an Eulerian circuit of the multigraph with $c(a)$ copies of each content, where $c + \rho c = m$ and m is the multiplicity vector of D_k . With $f = 2c - m$ one has $f \in \Lambda^- = \{f \in Z_1(D_k) : \rho f = -f\}$ inside the box $|f| \leq m$, and

$$\#\text{DS}(S, k) = \sum_f N_{\text{seq}}\left(\frac{m+f}{2}\right)$$

[1]. Here $(\rho z)(a) = z(\rho a)$ on 1-chains. A vertex is *palindromic* if $\bar{v} = v$, and a content is *self-complementary* if $\rho a = a$.

3 Inversions at inverted repeats

Theorem 1 (flips preserve the double-stranded spectrum [1, Prop. (flip)(a)]). *Let T be circular and w a k -mer with an occurrence of w and a later occurrence of $\text{rc}(w)$, so that, read from the first, $T = ZR$ with $Z = wZ'\text{rc}(w)$. Let $T' = \text{rc}(Z)R$. Then $\text{spec}_{k+1}(T') + \text{spec}_{k+1}(\text{rc } T') = \text{spec}_{k+1}(T) + \text{spec}_{k+1}(\text{rc } T)$.*

Proof. $\text{rc}(Z) = w \text{rc}(Z') \text{rc}(w)$ begins with w and ends with $\text{rc}(w)$, as Z does. A $(k+1)$ -mer of T that straddles a junction between Z and R meets Z in at most k letters, and those letters lie inside the flanking copy of w or $\text{rc}(w)$. So it is also a $(k+1)$ -mer of T' . The $(k+1)$ -mers inside Z are replaced by their reverse complements, so $\text{spec}(T') = \text{spec}(T) - \text{spec}(Z) + \text{rc spec}(Z)$. Applying rc and adding gives the claim. $\square$

	T	T + R	T + I	T + I + R
random cases whose closure is all of DS (of 24)	0	2	23	24
$\phi X174$, $k = 12$ (#DS = 10)	2	4	10	10
$\phi X174$, $k = 11$ (#DS = 864)	33	66	864	864

Table 1: Closures of S under the move families: T transpositions, R reverse complement, I inversions at inverted repeats. For $\phi X174$ the entries are the sizes of the closures. At $k = 12$ and $k = 11$ the full closure visits 8 and 218 lattice points, exactly the box points with connected support found by Bio 8 and Bio 12.

This is Bio 8’s Proposition on the inversion flip, part (a), and its proof is Bio 8’s. We repeat it only to fix notation; in Bio 8’s words, the move of this section is a *flip*.

On the double cover, the two copies w , $\text{rc}(w)$ are vertices v , $\bar{v}$. The segment Z is a path z of contents from v to $\bar{v}$, and ρz is again a path from v to $\bar{v}$, because ρ reverses paths and exchanges their ends. The inversion replaces z by ρz :

$$c' = c - z + \rho z, \quad f' = f - 2(1 - \rho)z. \quad (1)$$

The vector $(1 - \rho)z$ is a cycle, since z and ρz have the same ends, and it is ρ -antisymmetric. So it lies in Λ^- . An inversion changes the strand of every content of Z . A transposition changes no multiplicity, so $f' = f$.

Check (I) of `bio17_verify.py` tests Theorem 1 directly on strings. All 156 inversions at inverted repeats of the random test sequences preserve the double-stranded spectrum. So do all 2, 9 and 31 such inversions of $\phi X174$ itself at $k = 12$, 11 and 10. As a control, only 64 of 960 inversions of random, unflanked segments preserve it, and those are chance coincidences at small k .

4 Which moves generate the fibre

We represent a reconstruction as a cyclic sequence of contents and apply three families of moves:

- **T** (transpositions): at a vertex visited three times, exchange two of the returning loops; or, at two vertices $u \neq w$ visited alternately $u w u w$, exchange the two $u \rightarrow w$ segments;
- **I**: the inversions of Theorem 1, at every pair of visits to v and $\bar{v}$;
- **R**: global reverse complementation.

The closure of S under a family is computed by breadth-first search, and every state is checked to be a valid reconstruction.

Three counts agree. On twenty-four random sequences of length 16–33 with $k = 3$ –5, chosen so that at least two lattice points carry reconstructions, three quantities coincide in every case. The first is the set of circular strings with the double-stranded spectrum of S , found by exhaustive search. The second is Bio 8’s lattice sum. The third is the closure of S under T, I and R. The fibres range from 4 to 1 872 sequences (check (C)).

For $\phi X174$ these closures refine Bio 8’s Theorem on flips, which showed that at $k = 12$ and $k = 11$ the flips *alone*, without transpositions and without reverse complementation, already

connect the fibre [1, §7]. What transpositions add is the general case: Bio 8's Proposition (flip)(d) shows that a genome without inverted repeats admits no flip at all. Table 1 separates the contributions. Transpositions alone give the single-stranded fibre of S : N_{seq} at its own lattice point, 2 and 33 for ϕX174 . Reverse complementation doubles this. The inversions are what carry the closure across the lattice. At $k = 11$ they alone, with transpositions, reach all 864 reconstructions and all 218 lattice points with connected support. In one random case of twenty-four, T and I reach only part of the fibre and R is needed as well. By Bio 8's Proposition (flip)(c), reverse complementation is itself a flip only at a palindromic visit. We have not checked whether the exceptional case lacks one.

Conjecture 2 (double-stranded Ukkonen–Pevzner). *For every circular S and every k , $\text{DS}(S, k)$ is connected under transpositions at repeated k -mers, inversions at inverted repeats of length k , and global reverse complementation.*

Within one lattice point the conjecture is the classical theorem. The open part is the passage between lattice points: whether any two box points with connected support are joined by a chain of inversions that are realizable, i.e. whose flanking copies $v, \bar{v}$ are actually visited by the current circuit. Section 5 shows that nothing is lost at the level of the lattice.

5 The inversion lattice

By (1) the inversion vectors $(1 - \rho)z$, for all paths z from a vertex to its complement, generate a sublattice $M \subseteq \Lambda^-$. The differences of two such paths with the same ends are cycles, so M contains $(1 - \rho)Z_1$. The question is whether M , or already $(1 - \rho)Z_1$, is all of Λ^- . The group $\langle \rho \rangle \cong \mathbb{Z}/2$ acts, and the answer is a Tate-cohomology computation. Recall that for $\mathbb{Z}/2$ acting on a module A one has $\hat{H}^0(A) = \ker(1 - \rho)/\text{im}(1 + \rho)$ and $\hat{H}^1(A) = \ker(1 + \rho)/\text{im}(1 - \rho)$ [8]. In particular $\Lambda^-/(1 - \rho)Z_1 = \hat{H}^1(Z_1)$.

Theorem 3 (the Tate dichotomy). *Let D_k be connected. Then*

$$\Lambda^-/(1 - \rho)Z_1 \cong \begin{cases} \mathbb{Z}/2 & \text{if } D_k \text{ has no palindromic vertex and no self-complementary content,} \\ 0 & \text{otherwise.} \end{cases}$$

In every case $M = \Lambda^-$. In the first case the non-trivial class is represented by $(1 - \rho)p$ for any path p from a vertex v to $\bar{v}$.

Proof. Let C_1 be the free abelian group on the contents and C_0 that on the vertices, with $\partial a = h(a) - t(a)$. Because ρa runs from $\bar{h}$ to $\bar{t}$, one has $\partial \rho = -\rho \partial$. So ∂ is equivariant when C_0 carries the twisted action $\sigma = -\rho$. Let $B_0 = \partial C_1$. Since D_k is connected, B_0 is the group of 0-chains of total degree 0. The sequence $0 \rightarrow Z_1 \rightarrow C_1 \rightarrow B_0 \rightarrow 0$ is exact and equivariant, and its long exact sequence in Tate cohomology reads

$$\hat{H}^0(C_1) \longrightarrow \hat{H}^0(B_0) \xrightarrow{\delta} \hat{H}^1(Z_1) \longrightarrow \hat{H}^1(C_1).$$

C_1 . The contents fall into free orbits $\{a, \rho a\}$, each a copy of $\mathbb{Z}[\mathbb{Z}/2]$ with vanishing Tate cohomology, and self-complementary contents, each a copy of the trivial module $\mathbb{Z}$, with $\hat{H}^0 = \mathbb{Z}/2$ and $\hat{H}^1 = 0$. So $\hat{H}^1(C_1) = 0$, and $\hat{H}^0(C_1) = (\mathbb{Z}/2)^{\#\text{self-compl.}}$, generated by the classes of the self-complementary contents.

double cover	covers	$[\Lambda^- : (1 - \rho)Z_1]$	$[\Lambda^- : M]$
no palindromic vertex, no self-complementary content	60	2 in all 60	1
self-complementary content, no palindromic vertex	241	1	1
palindromic vertex, no self-complementary content	299	1	1
ϕX174 , $k = 10$ (3 palindromic vertices)	1	1	1
ϕX174 , $k = 11$ (1 self-complementary content)	1	1	1
ϕX174 , $k = 12$ (1 palindromic vertex)	1	1	1

Table 2: Indices computed by Smith normal form on 600 random connected double covers (lengths 14–59, $k = 2$ –7; 36 disconnected covers skipped) and on ϕX174 . The ranks of both lattices always equal $\text{rank } \Lambda^-$: 48, 12 and 4 for ϕX174 .

B_0 . $\ker(1 - \sigma)$ consists of the x with $x(\bar{v}) = -x(v)$. Such an x vanishes at palindromic vertices, and is spanned by the $e_v - e_{\bar{v}}$. Now $\text{im}(1 + \sigma)$ consists of the $y - \rho y$ with $y \in B_0$. If p is a palindromic vertex, then $e_v - e_{\bar{v}} = (1 - \rho)(e_v - e_p)$ with $e_v - e_p \in B_0$, so $\hat{H}^0(B_0) = 0$. If there is none, any y with $(1 - \rho)y = e_v - e_{\bar{v}}$ has the form $e_v + s$ with $\rho s = s$, and s has even total degree, because its values pair up over orbits of size two. So $y \notin B_0$. Hence $\hat{H}^0(B_0) = \mathbb{Z}/2$, generated by the common class of all $e_v - e_{\bar{v}}$; they differ by $(1 - \rho)(e_v - e_u)$ with $e_v - e_u \in B_0$.

The map. A self-complementary content $a: t \rightarrow h$ has $h = \bar{t}$, so $\partial a = e_{\bar{t}} - e_t$, the generator of $\hat{H}^0(B_0)$. So if a self-complementary content exists, $\hat{H}^0(C_1) \rightarrow \hat{H}^0(B_0)$ is onto and $\hat{H}^1(Z_1) = 0$. If none exists and there is no palindromic vertex, $\hat{H}^1(Z_1) \cong \hat{H}^0(B_0) = \mathbb{Z}/2$. If there is a palindromic vertex, $\hat{H}^0(B_0) = 0$ and again $\hat{H}^1(Z_1) = 0$.

The representative. The connecting map sends the class of $x = e_{\bar{v}} - e_v$ to the class of $z - \rho z$, for any $z \in C_1$ with $\partial z = x$: then $\partial(\rho z) = -\rho x = x$, so $z - \rho z \in Z_1$. A path p from v to $\bar{v}$ is such a z . So $(1 - \rho)p$ represents the non-trivial class, and $M = \Lambda^-$. $\square$

Table 2 confirms the theorem exactly on every cover tested. The dichotomy has a transparent meaning. A palindromic vertex, or a self-complementary content, is a place where the two strands touch. Without one, flipping closed cycles, $(1 - \rho)Z_1$, can only reach a sublattice of index 2, and the missing half needs an open path from a word to its complement: a genuine inversion.

6 ϕX174 at $k = 10$

At $k = 10$ the fibre has 31 925 753 246 212 elements on $1\,240\,843\,950 \times 2$ box points [3]. It cannot be closed, but it can be walked. The genome carries 31 inversions at inverted repeats of length 10. The count on the cover is 32. The extra one comes from a copy of $\text{rc}(w)$ at position 4482 and a copy of w at 4484. Their two flanks overlap, and the string-level count excludes that case. One inversion from ϕX174 reaches 23 other lattice points. Two random walks of 20 000 moves each, one without and one with global reverse complementation, stay inside the fibre at every step: each state is checked to be a valid reconstruction, and the final one also at the level of strings. They visit 4 907 and 5 049 distinct lattice points, and none of the sampled sequences repeats. That is consistent with Conjecture 2 but does not test it: at this scale the question is not whether the walk can move, but whether it can reach every one of the $454\,551\,605 \times 2$ points with connected support.

7 Biological reading

Theorem 1 says exactly which rearrangements double-stranded k -mer data cannot see: transpositions between direct repeats, and inversions between inverted repeats, each at least k long. These are the two classical products of non-allelic homologous recombination: deletions, duplications and translocations between direct repeats, and inversions between inverted ones [9]. The first class preserves the content of one strand and the second only the content of both. Any strand-agnostic encoder that depends on k -mer content alone is blind to both. That includes a reverse-complement-invariant local DNA encoder of window $k + 1$, in the sense of our companion note on spectral fibres [4]. A short-read assembler faces the same limit. The inversion lattice being all of Λ^- means that, algebraically, these two mechanisms already account for every direction in which the double-stranded fibre extends.

8 On the proposed inversion-group programme

This paper was specified with a different programme: a group G_{inv} generated by adjacent exchanges and segment inversions, embedded in an Artin–Tits group, shown to be Garside, with the minimum inversion distance read off a Garside normal form. We set it aside for three reasons.

- The group generated by signed permutations of n oriented segments is the finite hyperoctahedral group B_n , of order $2^n n!$. It is a Coxeter group, not an Artin group, and it does not embed in one as the specification intended. Its word problem is trivial.
- The minimum number of inversions between two signed orders, the reversal distance, is computed in polynomial time by Hannenhalli and Pevzner [7]. Word length in B_n with respect to its Coxeter generators is a different quantity, and it does not model inversions of arbitrary segments.
- The object that is new, and that belongs to algebraic genomics, is not the group of all inversions but the inversions that preserve the double-stranded spectrum. Those are the ones flanked by inverted repeats, and they are what this note studies.

Ancillary file

`bio17_verify.py` (standard library; imports `ds_fermion.py` of Bio 13) runs checks (I), (C), (G), (L), (H) and (W). These are the inversion theorem on strings; the three counts; the closures; the inversion lattice on ϕX174 and random covers; the Tate dichotomy on 600 random covers; and the walks at $k = 10$. The whole battery takes under two minutes.

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